

Caringbah High School



2011 Trial Higher School Certificate Examination

Mathematics

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks - 120

- Attempt Questions 1 -10
- All questions are of equal value

Total marks – 120

Attempt Questions 1–10

All questions are of equal value

Start each question in a SEPARATE booklet. Extra booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- (a) Evaluate $\sqrt{\frac{4^2+11^2}{321-11^2}}$ correct to three significant figures. 2
- (b) Find a primitive of $4 + \sec^2 x$. 2
- (c) Factorise $x^3 + 27$. 2
- (d) Solve $|x - 5| = 8$. 2
- (e) Simplify $\frac{x}{x^2 - 9} + \frac{3}{x + 3}$. 2
- (f) Solve $4x = x^2$. 2

Question 2 (12 marks) Use a SEPARATE writing booklet.

- (a) Differentiate with respect to x :
- (i) $x^5 + 7$ 2
- (ii) $\frac{x^2}{x+1}$ 2
- (iii) $x \cos x$ 2

Question 2 (continued)

(b) Find

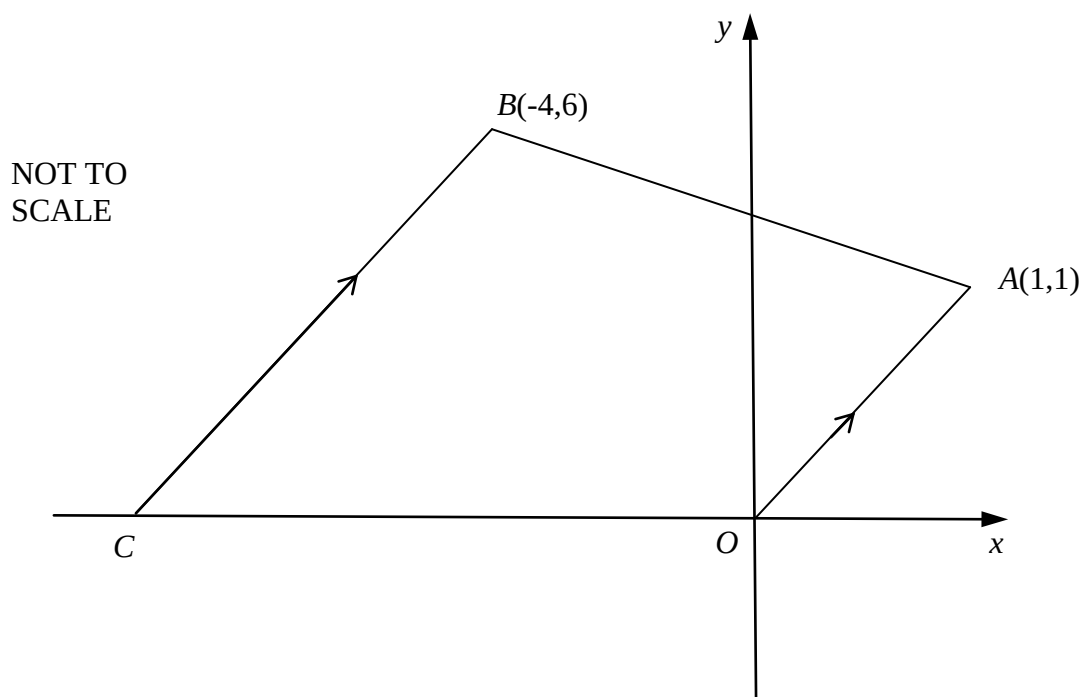
(i) $\int \frac{12x^3}{x^4 + 2} dx$ 2

(ii) $\int_0^1 (e^{5x} - 1) dx$ 2

(c) Find the equation of the tangent to $y = e^{2x}$ at the point $(2, e^4)$. 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

(a)



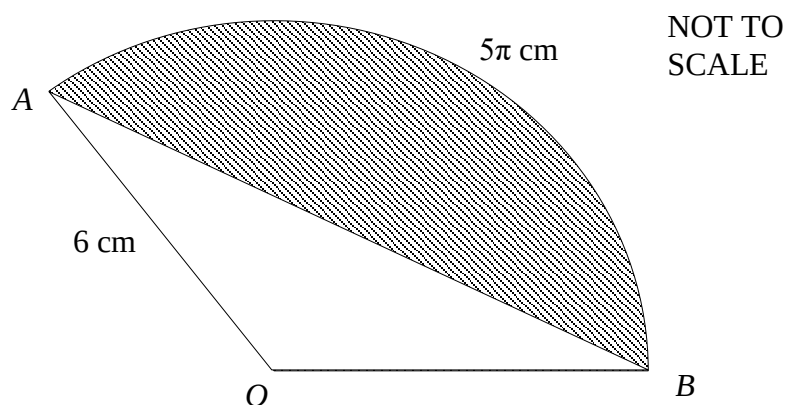
In the diagram, $OABC$ is a trapezium with $OA \parallel CB$. The coordinates of O , A and B are $(0,0)$, $(1,1)$ and $(-4,6)$ respectively.

- | | | |
|-------|---|---|
| (i) | Calculate the length OA . | 1 |
| (ii) | Write down the gradient of line OA . | 1 |
| (iii) | What is the size of $\angle AOC$? | 1 |
| (iv) | Find the equation of the line BC , and hence find the coordinates of C . | 2 |
| (v) | Show that the perpendicular distance from O to the line BC is $5\sqrt{2}$. | 2 |
| (vi) | Hence, or otherwise, calculate the area of the trapezium $OABC$. | 2 |
| (b) | The lengths of the sides of a triangle are 8 cm, 9 cm and 14 cm. Find the size of the angle opposite the smallest side. | 2 |

(c)	Evaluate $\sum_{n=2}^4 (1-3n)$	1
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Question 4 (12 marks) Use a SEPARATE writing booklet.

(a)



AOB is a sector of a circle, centre *O* and radius 6 cm. The length of the arc *AB* is 5π cm.

- | | | |
|------|---|---|
| (i) | Find the exact size of $\angle AOB$ | 1 |
| (ii) | Calculate the exact area of the shaded segment. | 2 |
- (b) Consider the function $f(x) = 3x^2 - x^3$.
- | | | |
|-------|---|---|
| (i) | Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. | 3 |
| (ii) | Sketch the curve showing where it meets the axes. | 2 |
| (iii) | Find the values for which the curve is concave down. | 2 |
- (c) Sally invests \$3000 in a term deposit that earns 6.5% per annum compounded annually. What is the value of her investment at the end of 15 years?
- | | | |
|--|--|---|
| | | 2 |
|--|--|---|

Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) Madison is learning to drive. Her first lesson is 10 minutes long. Her second lesson is 15 minutes long. Each subsequent lesson is 5 minutes longer than the previous lesson.
- (i) How long will Madison's fifteenth lesson be? 1
- (ii) How many hours of lessons will Madison have completed after her fifteenth lesson? 2
- (iii) During which lesson will Madison have completed a total of 40 hours of driving lessons? 2
- (b) Find the values of m for which the expression below is always positive. 2
- $$x^2 + 2mx + (3m - 2)$$
- (c) Find the amplitude and period if $y = -3\cos \pi x$. 2
- (d) (i) Differentiate $\log_e(\sin x)$ 1
- (ii) Hence, or otherwise, find the exact value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot x \, dx$ 2

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) Find all values of θ , where $0^\circ \leq \theta \leq 360^\circ$, that satisfy the equation $\cos \theta - \frac{2}{5} = 0$. 2
- Give your answer(s) to the nearest degree.
- (b) Solve $|x - 2| > 5$ and graph your solution on the number line. 2

Question 6 (continued)

- (c) (i) Find the limiting sum of the geometric series 2

$$3 + \frac{3}{\sqrt{3}+1} + \frac{3}{(\sqrt{3}+1)^2} + \dots$$

- (ii) Explain why the geometric series 1

$$3 + \frac{3}{\sqrt{3}-1} + \frac{3}{(\sqrt{3}-1)^2} + \dots$$

does NOT have a limiting sum.

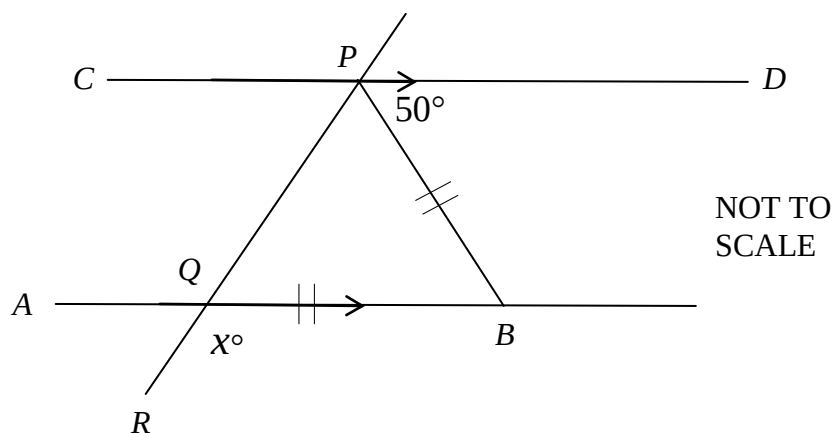
- (d) A council worker accidentally spread a toxic chemical on a local soccer field. The concentration of the chemical in the soil was initially measured at 4 kg/ha. One year later the concentration was found to be 2.6 kg/ha. It is known that the concentration, C , is given by $C = C_0 e^{-kt}$, where C_0 and k are constants, and t is measured in years.

- (i) Evaluate C_0 and k . 3

- (ii) It is safe to use the soccer field when the concentration is below 0.1 kg/ha. How long must the soccer players wait after the accident before the soccer field can be used? Give your answer in years, correct to one decimal place. 2

Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a)



3

In the diagram, CD is parallel to AB , $PB = QB$, $\angle BPD = 50^\circ$ and $\angle BQR = x^\circ$. Copy or trace this diagram.

Find the value of x , giving complete reasons.

Question 7 (continued)

- (b) Let α and β be the roots of the equation

$$x^2 - 5x + 2 = 0$$

Find the values of;

- (b) (i) $\alpha + \beta$ **1**

- (ii) $\alpha\beta$ **1**

- (iii) $(\alpha + 1)(\beta + 1)$ **2**

- (c) The equation of a parabola is $y = \frac{x^2}{8} - 3$

- (i) Find the coordinates of the vertex of the parabola. **2**

- (ii) Find the equation of the directrix of the parabola. **1**

- (iii) Sketch the curve clearly labelling the vertex and directrix. **2**

Question 8 (12 marks) Use a SEPARATE writing booklet.

- (a) Use Simpson's rule, with three function values to find an **2**

approximation for $\int_{0.5}^{1.5} (\log_e x)^3 dx$.

Give your answer correct to three decimal places.

- (b) (i) Write down the discriminant of $2x^2 + (k - 2)x + 8$, where k is a constant. **1**

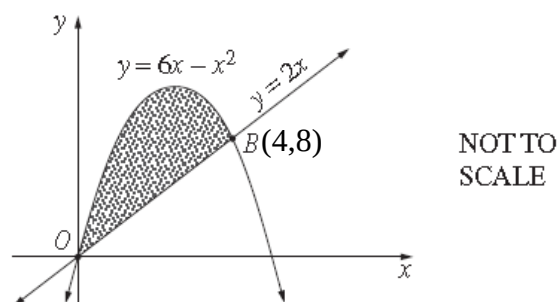
- (ii) Hence, or otherwise, find the values of k for which the parabola $y = 2x^2 + kx + 9$ does not intersect the line $y = 2x + 1$. **2**

Question 8 (continued)

- (c) During a storm, water flows into a 7000 litre tank at a rate of $\frac{dV}{dt}$ litres per minute, where $\frac{dV}{dt} = 120 + 26t - t^2$ and t is the time in minutes since the storm began.
- (i) At what time is the tank filling at twice the initial rate? 2
- (ii) Find the volume of water that has flowed into the tank since the start of the storm as a function of t . 1
- (iii) Initially, the tank contains 1500 litres of water. When the storm finishes, 30 minutes after it began, the tank is overflowing. How many litres of water have been lost? 2
- (d) Solve the equation $2 \ln x = \ln(7x - 12)$ 2

Question 9 (12 marks) Use a SEPARATE writing booklet.

- (a) 3

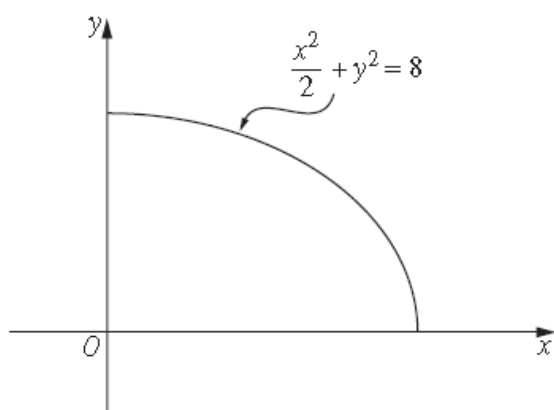


The graphs of $y = 2x$ and $y = 6x - x^2$ intersect at the origin and point $B(4,8)$.

Find the shaded area bounded by $y = 2x$ and $y = 6x - x^2$.

Question 9 (continued)

(b)



The part of the curve $\frac{x^2}{2} + y^2 = 8$ that lies in the first quadrant is rotated about the x -axis. Find the volume of the solid of revolution.

3

- (c) Mr Smith borrows \$80 000 to purchase a new car. The interest rate is calculated monthly at the rate of 1% per month, and is compounded each month.

Mr Smith intends to repay the loan, with interest, in two equal annual instalments of \$ M at the end of the first and second years.

- | | | |
|-------|---|----------|
| (i) | How much does Mr Smith owe at the end of the first month? | 1 |
| (ii) | Write an expression involving M for the total amount owed by Mr Smith after 12 months, just after the first instalment of \$ M has been paid. | 2 |
| (iii) | Find an expression for the amount owed at the end of the second year and deduce that | 2 |

$$M = \frac{80000 \times (1.01)^{24}}{(1.01)^{12} + 1}$$

(iv) What is the total interest over the two-year period?

1

Question 10 (12 marks) Use a SEPARATE writing booklet.

- (a) Solve the following equation for x : 2

$$2e^{2x} - e^x = 0$$

- (b) Show that $f(x) = \frac{x^4 - 8}{x^3}$ is an odd function. 1

- (c) Let $f(x) = \sqrt{25 - x^2}$

- (i) Copy the following table of values into your writing booklet and supply the missing values. 1

x	0	1	2	3	4	5
$f(x)$	5.000		4.583			0.000

- (ii) Use these six values of the function and the trapezoidal rule to find the approximate value of 2

$$\int_0^5 \sqrt{25 - x^2} dx$$

- (iii) Draw the graph of $x^2 + y^2 = 25$ and shade the region whose area is represented by the integral 2

$$\int_0^5 \sqrt{25 - x^2} dx$$

- (iv) Use your answer to part (iii) to explain why the exact value of the integral is $\frac{25\pi}{4}$. 2

- (v) Use your answers to part (ii) and part (iv) to find an approximate value of π . 2

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1

- (a) 0.828
 (b) $4x + \tan x + c$
 (c) $(x+3)(x^2 - 3x + 9)$
 (d) $x = -3$ or 13
 (e) $\frac{x}{(x-3)(x+3)} + \frac{3(x-3)}{(x+3)(x-3)} = \frac{4x-9}{(x-3)(x+3)}$
 (f) $x^2 - 4x = 0$
 $x(x-4) = 0 \Rightarrow x = 0$ or 4

Q2 (a) (i) $5x^4$

$$(ii) \frac{2x(x+1) - 1 \cdot x^2}{(x+1)^2} = \frac{x^2 + 2x}{(x+1)^2}$$

$$= \frac{x(x+2)}{(x+1)^2}$$

(iii) $\cos x - x \sin x$

(b) (i) $3 \int \frac{4x^3}{x^4 + 2} dx = 3 \ln(x^4 + 2) + c$

$$(ii) \left[\frac{e^{5x}}{5} - x \right]_0^1 = \frac{e^5}{5} - 1 - \left(\frac{1}{5} - 0 \right)$$

$$= \frac{e^5}{5} - \frac{6}{5}$$

$$= \frac{e^5 - 6}{5}$$

(c) $y' = 2e^{2x} \Rightarrow m_T(2) = 2e^4$

$$\therefore y - e^4 = 2e^4(x - 2)$$

$$y = 2e^4x - 4e^4 + e^4$$

$$\therefore \text{Tangent is } y = 2e^4x - 3e^4$$

Q3 (a) (i) $OA = \sqrt{(1-0)^2 + (1-0)^2} = \sqrt{2}$

$$(ii) m_{OA} = \frac{1-0}{1-0}$$

$$= 1$$

(iii) $\angle AOC = 135^\circ$

(iv) $m_{BC} = 1 \quad B = (-4, 6)$

$$y - 6 = 1(x - (-4))$$

$$y = x + 4 + 6$$

line BC $\equiv y = x + 10$

at C $y = 0 \therefore x = -10$

(v) line BC $\equiv x - y + 10 = 0$

$$\therefore D_{BC} = \frac{|0 \cdot 1 - 0 \cdot (-10)|}{\sqrt{1^2 + 1^2}}$$

$$= \frac{|10|}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{10\sqrt{2}}{2}$$

$$= 5\sqrt{2}$$

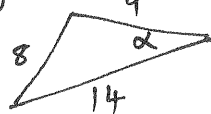
$$(vi) \text{Need } d_{BC} = \sqrt{(-10 - (-4))^2 + (6 - 0)^2}$$

$$= \sqrt{36 + 36}$$

$$= 6\sqrt{2}$$

$$\therefore \text{Area}_{OABC} = \frac{1}{2} \cdot 5\sqrt{2} \cdot (6\sqrt{2} + \sqrt{2})$$

$$= 35\sqrt{2}$$



$$\cos \alpha = \frac{9^2 + 14^2 - 8^2}{2 \cdot 9 \cdot 14}$$

$$\therefore \alpha \approx 32.3025^\circ$$

$$\approx 32^\circ 18'$$

$$(c) \text{Sum} = (1-12) + (1-9) + (1-6)$$

$$= -11 - 8 - 5$$

$$= -24$$

Q4 (a) (i) $l = r\theta$

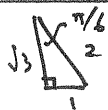
$$5\pi = 6\theta$$

$$\theta = \frac{5\pi}{6}$$

$$(ii) \text{Area} = \frac{1}{2} 6^2 \left(\frac{5\pi}{6} - \sin \frac{5\pi}{6} \right)$$

$$= 15\pi - 18 \sin \frac{\pi}{6}$$

Q4(a)(ii) (cont'd) $= 15\pi - 18 \cdot \frac{1}{2}$
 $= 15\pi - 9$



(b) (i) $f'(x) = 6x - 3x^2$
 $= 3x(2 - x) = 0$
 when $x = 0$ or 2

If $x = 0, y = 0 \Rightarrow (0, 0)$

If $x = 2, y = 4 \Rightarrow (2, 4)$

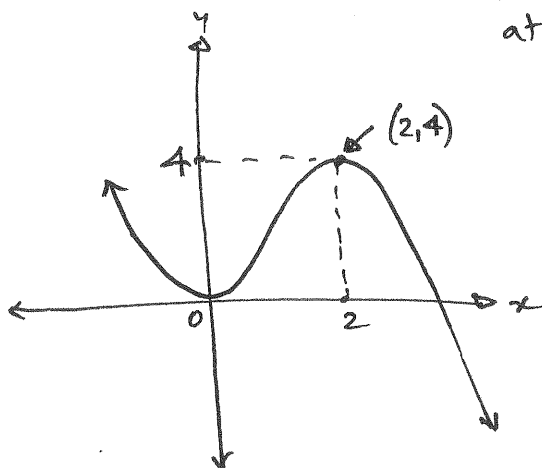
To get nature consider $f''(x)$

$f''(x) = 6 - 6x$
 $= 6(1 - x)$

$f''(0) = 6 > 0 \Rightarrow \cup \Rightarrow$ Minimum
 Turning point at $(0, 0)$

$f''(2) = -6 < 0 \Rightarrow \cap \Rightarrow$ Maximum
 Turning point at $(2, 4)$

(ii)



(iii) concave down $\Rightarrow f''(x) < 0 \Rightarrow 6 - 6x < 0$
 $6x > 6$
 $x > 1$

(c) Use $A = P(1+r)^n$
 $= \$3000(1.065)^{15}$
 $= \$7715.52$

Q5(a)(i) $a = 10, d = 5, T_{15} = ?$

Use $T_n = a + (n-1)d$

$T_{15} = 10 + 14 \times 5$
 $= 80$ minutes

(ii) $S_{15} = ?$ Use $S_n = \frac{n}{2} \{a + l\}$
 $= \frac{15}{2} (10 + 80)$

$= 675$ minutes
 $= 11.25$ hours.

(iii) $S_n = 40 \times 60, n = ?$
 Use $S_n = \frac{n}{2} (2a + (n-1)d)$

$40 \times 60 = \frac{n}{2} (20 + 5n - 5)$

$4800 = 15n + 5n^2$

$n^2 + 3n - 960 = 0$

$n = \frac{-3 \pm \sqrt{9 + 4 \times 960}}{2}$

$n = 29.5$

$\therefore$ During the 30th lesson

(b) Always positive $\Rightarrow \Delta > 0$

$\therefore b^2 - 4ac = (2m)^2 - 4(3m-2) > 0$

$4m^2 - 12m + 8 > 0$

$m^2 - 3m + 2 > 0$

$(m-2)(m-1) > 0$

$\therefore m < 1$ or $m > 2$

(c) amplitude = 3
 period = $\frac{2\pi}{\pi}$
 $= 2$

(d)(i) $\frac{\cos x}{\sin x} = \cot x$

(ii) $\left[\ln(\sin x) \right]_{\pi/4}^{\pi/2}$
 $= \ln(\sin \frac{\pi}{2}) - \ln(\sin \frac{\pi}{4})$
 $= \ln(1) - \ln(\frac{1}{\sqrt{2}})$
 $= \ln \sqrt{2}$

Q6(a) $\cos \theta = \frac{2}{3}$
 $\theta = 66^\circ$ or 294°

(b) $x - 2 > 5$ or $-(x - 2) > 5$
 $x > 7$ or $x < -3$



(c)(i) $a = 3, r = \frac{1}{\sqrt{3} + 1}$

$S_{\infty} = \frac{3}{1 - \frac{1}{\sqrt{3} + 1}}$

$$= \frac{3(\sqrt{3}+1)}{\sqrt{3}+1-1} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \sqrt{3}(\sqrt{3}+1)$$

$$= 3 + \sqrt{3}$$

$$(ii) \quad r = \frac{1}{(\sqrt{3}-1)} \cdot \left(\frac{\sqrt{3}+1}{\sqrt{3}+1} \right)$$

$$= \frac{\sqrt{3}+1}{3-1}$$

$$= \frac{1}{2}(1+\sqrt{3})$$

$$\approx 1.366 > 1 \Rightarrow \text{no limiting sum}$$

$$(d)(i) \quad C_0 = 4$$

$$\text{and } 2.6 = 4 \times e^{-kt}$$

$$\frac{2.6}{4} = e^{-k}$$

$$k = -\ln\left(\frac{2.6}{4}\right) \div 0.4307829161$$

$$(ii) \quad 0.1 = 4 \times e^{-kt}$$

$$e^{-kt} = \frac{0.1}{4}$$

$$-kt = \ln\left(\frac{0.1}{4}\right)$$

$$t = -\frac{\ln\left(\frac{0.1}{4}\right)}{k}$$

$$\div 8.563198113 \text{ yrs}$$

$$\div 8.6 \text{ yrs}$$

$$Q7 (a) \quad \triangle BPQ \text{ is isosceles } (BP = BQ)$$

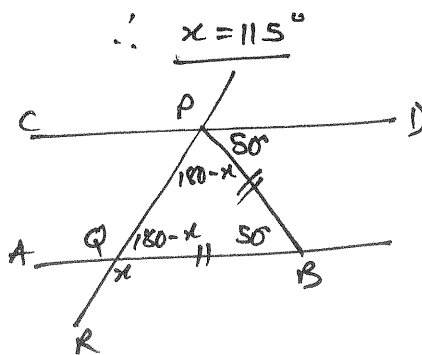
$$\angle BQP = 180^\circ - x \quad (\text{angle sum straight line } RQP)$$

$$\angle BPQ = \angle BQP \quad (\text{base angles isosceles triangle } BPQ)$$

$$= 180^\circ - x$$

$$\angle QBP = \angle BPD = 50^\circ \quad (\text{alternate angles, } PD \parallel QB)$$

$$\therefore 180 - x + 180 - x + 50 = 180 \quad (\text{angle sum, } \triangle BPQ)$$



$$(b)(i) \quad \alpha + \beta = \frac{-b}{a} = 5$$

$$(ii) \quad \alpha\beta = \frac{c}{a} = 2$$

$$(iii) \quad (\alpha+1)(\beta+1) = \alpha\beta + \alpha + \beta + 1$$

$$= 2 + 5 + 1$$

$$= 8$$

$$(c)(i) \quad 8y = x^2 - 24$$

$$x^2 = 8y + 24$$

$$= 8(y+3)$$

$$(x-0)^2 = 4 \times 2 \times (y+3)$$

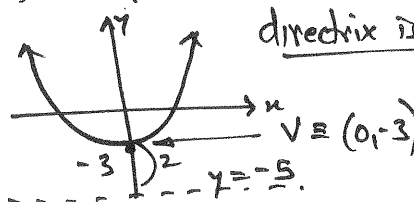
$$\therefore \text{Vertex} = (0, -3)$$

$$(ii)$$

$$a=2$$

$$\text{directrix is } y = -5$$

$$(iii)$$



$$Q8 (a)$$

$$\int_{0.5}^{1.5} (\log_e x)^3 dx \div \frac{1}{6} \left\{ (\ln(0.5))^3 + 4(\ln(1))^3 + (\ln(1.5))^3 \right\}$$

$$\div -0.044$$

$$(b)(i) \quad \Delta \equiv (k-2)^2 - 4 \cdot 2 \cdot 8$$

$$= k^2 - 4k - 60$$

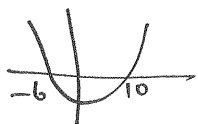
$$= (k-10)(k+6)$$

$$(ii) \quad 2x^2 + kx + 9 = 2x + 1$$

$$2x^2 + (k-2)x + 8 = 0$$

$$\text{no solutions} \Rightarrow \text{no intersections} \Rightarrow \Delta < 0$$

$$\therefore (k-10)(k+6) < 0$$



$$\therefore -6 < k < 10$$

(c) (i) initial rate $\Rightarrow t=0$

$$\therefore \frac{dv}{dt} = 120 + 0 - 0 = 120 \text{ L/min}$$

$$\therefore 240 = 120 + 26t - t^2$$

$$t^2 - 26t + 120 = 0$$

$$(t-6)(t-20) = 0$$

$$\therefore t = 6 \text{ or } 20 \text{ min}$$

(ii) $V = \int (120 + 26t - t^2) dt$

$$V = 120t + 13t^2 - \frac{t^3}{3} + c$$

(iii) $t=0, V=1500$

$$\therefore 1500 = 0 + 0 - 0 + c$$

$$\therefore V = 1500 + 120t + 13t^2 - \frac{t^3}{3}$$

$$V(20) = 1500 + 120 \times 20 + 13 \times 20^2 - \frac{20^3}{3}$$

$$= 7800 \Rightarrow \underline{800 \text{ L lost}}$$

(d) $\ln(x^2) = \ln(7x-12)$

$$\therefore x^2 = 7x-12$$

$$x^2 - 7x + 12 = 0$$

$$(x-3)(x-4) = 0$$

$$\therefore x = 3 \text{ or } 4$$

Q9 (a) Area = $\int_0^4 (6x - x^2 - 2x) dx$

$$= \int_0^4 (4x - x^2) dx$$

$$= \left[2x^2 - \frac{x^3}{3} \right]_0^4$$

$$= 32 - \frac{64}{3} - (0-0)$$

$$= \underline{\underline{\frac{32}{3}}}$$

(b) $V_x = \pi \int_0^4 \left(8 - \frac{x^2}{2} \right) dx$

$$= \pi \left[8x - \frac{x^3}{6} \right]_0^4$$

$$= \pi \left(32 - \frac{64}{6} - (0-0) \right)$$

$$= \underline{\underline{\frac{64\pi}{3}}}$$

(c) let Amount owing after n months be AO_n

(i) $\therefore AO_1 = \$80000 + 1\% \text{ of } \80000

$$= \underline{1.01 \times \$80000}$$

(ii) $AO_2 = 1.01 \times AO_1$

$$= 1.01^2 \times \$80000$$

$$= 1.01$$

$$\therefore AO_{12} = \underline{1.01^{12} \times \$80000 - \$M = Y_1}$$

(iii) $AO_{13} = 1.01 \times Y_1$

$$AO_{14} = 1.01^2 \times Y_1$$

$$AO_{24} = 1.01^{12} \times Y_1 - \$M$$

$$= 1.01^{12} (1.01^{12} \times \$80000 - \$M) - \$M$$

as $AO_{24} = 0$

then $1.01^{24} \times \$80000 = 1.01^{12} M + M$

$$= M(1 + 1.01^{12})$$

$$\therefore M = \underline{\underline{\frac{1.01^{24} \times \$80000}{1 + 1.01^{12}}}}$$

(iv) $\therefore M = \$47,760.756$

$$\therefore \text{total paid back} = \$95,521.51$$

$$\therefore \text{total interest} = \underline{\underline{\$15,521.51}}$$

Q10 (a) let $u = e^x$, $\therefore$ solve $2u^2 - u = 0$

$$u(2u-1) = 0$$

$$u = 0 \text{ or } u = \frac{1}{2}$$

$$e^x \neq 0 \therefore e^x = \frac{1}{2}$$

$$\therefore x = \underline{\underline{\ln(0.5)}}$$

(b) $f(-x) = \frac{(-x)^4 - 8}{(-x)^3}$

$$= \frac{x^4 - 8}{-x^3}$$

$$= -\left(\frac{x^4 - 8}{x^3}\right)$$

$$= -f(x) \Rightarrow f(x) \text{ is an odd function}$$

(c)

(i)

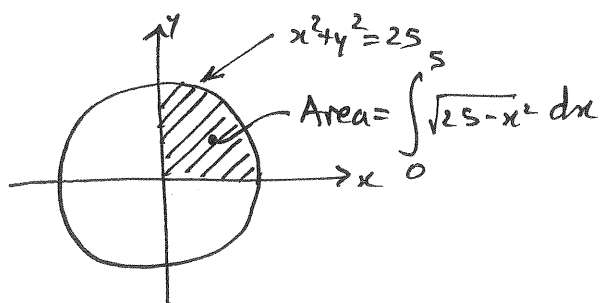
x	0	1	2	3	4	5
$f(x)$	5	$\sqrt{24}$	4.583	4	3	0

$$\sqrt{24} \doteq 4.899$$

(ii) $\int_0^5 \sqrt{25-x^2} dx \doteq \frac{1}{2} \left\{ 5 + 2 \times (\sqrt{24} + 4.583 + 4 + 3) + 0 \right\}$

$$\doteq \underline{18.98197949}$$

(iii)



(iv) $\int_0^5 \sqrt{25-x^2} dx = \text{area of } \frac{1}{4} \text{ of a circle}$
 centre (0,0), radius 5 units

$$= \frac{\pi \cdot 5^2}{4}$$

$$= \underline{\underline{\frac{25\pi}{4}}}$$

(v)

$$\therefore \frac{25\pi}{4} \doteq 18.98 \dots$$

$$\therefore \pi \doteq \frac{4}{25} \times 18.98 \dots$$

$$\doteq \underline{\underline{3.037116718}}$$